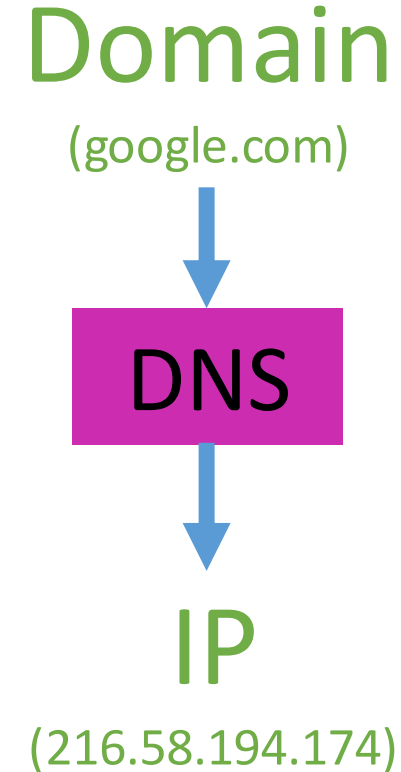


Domain Name System Forensics



Domain Name System (DNS)

- **Motivation:** Every device is identified by an IP address, which is a unique numerical identifier.
- **Problem:** However, IP addresses are numbers and can be difficult for people to remember and use.
- **Solution**
 - Each device has a human-readable string (domain name). Each domain name corresponds to an IP address
 - DNS translates domain names into IP addresses and vice versa.



DNS like a phone book for the internet

- Just like how you can look up your friend's phone number in a phone book to call them
 - your computer or tablet uses DNS to look up the address of a website you want to visit on the internet.
- When you type in the name of a website like YouTube
 - your computer or tablet sends a message to a special computer called a DNS server.
 - The DNS server looks up the address of YouTube and sends it back to your computer or tablet.
 - Then, your computer or tablet uses that address to find YouTube and show you all your favorite videos!
- So, the next time you use the internet, remember that DNS is working behind the scenes to make it all possible!



google.com
172.217.19.142



DNS Server
8.8.8.8

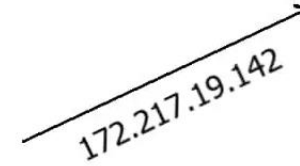


1. What is the IP of google.com?

google.com
172.217.19.142

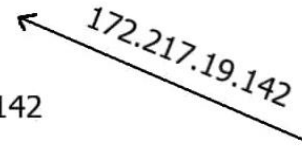
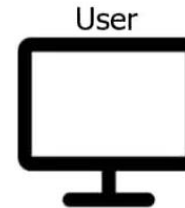


DNS Server
8.8.8.8



2. Here is the IP of google.com

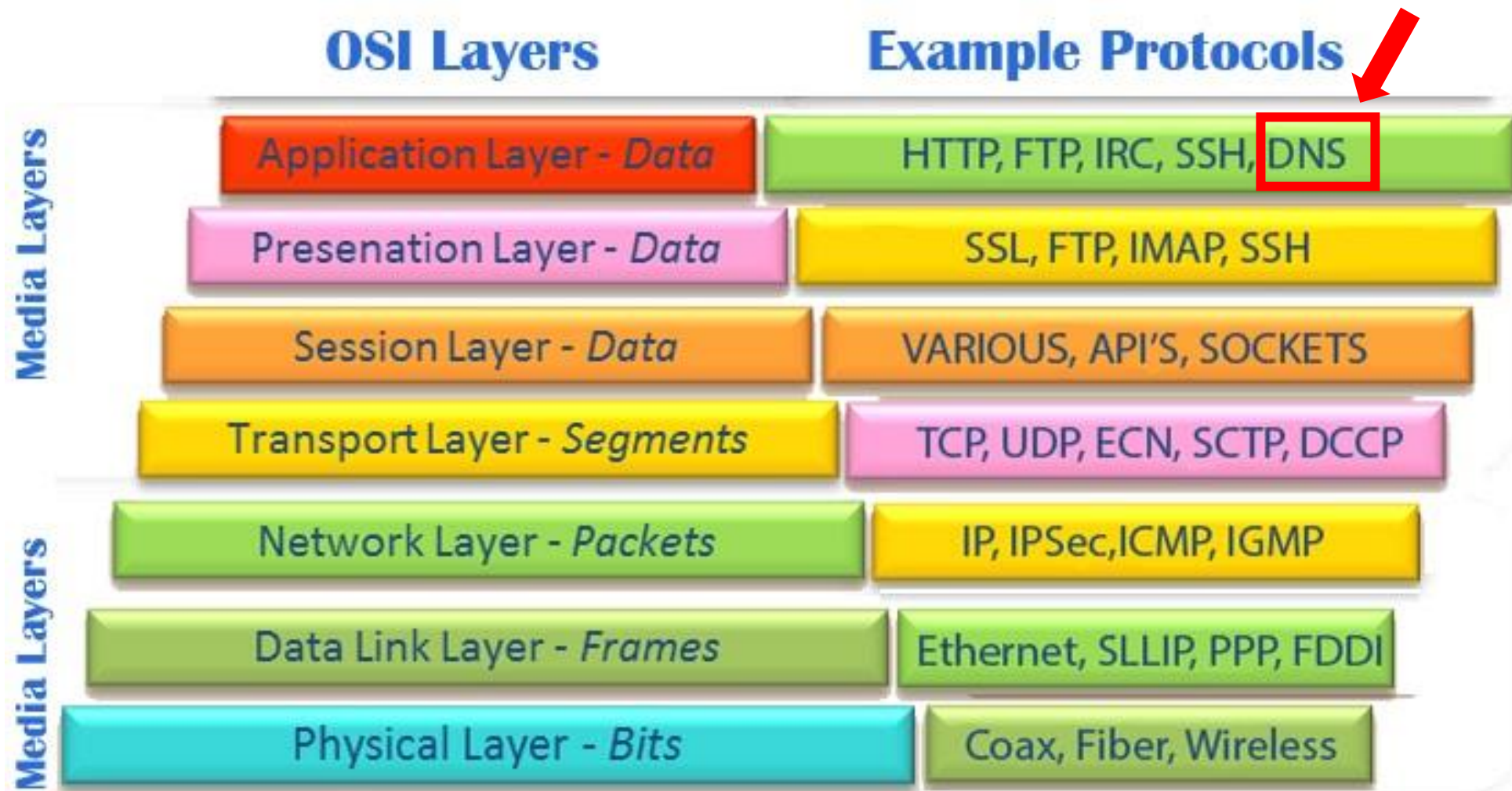
google.com
172.217.19.142



DNS Server
8.8.8.8

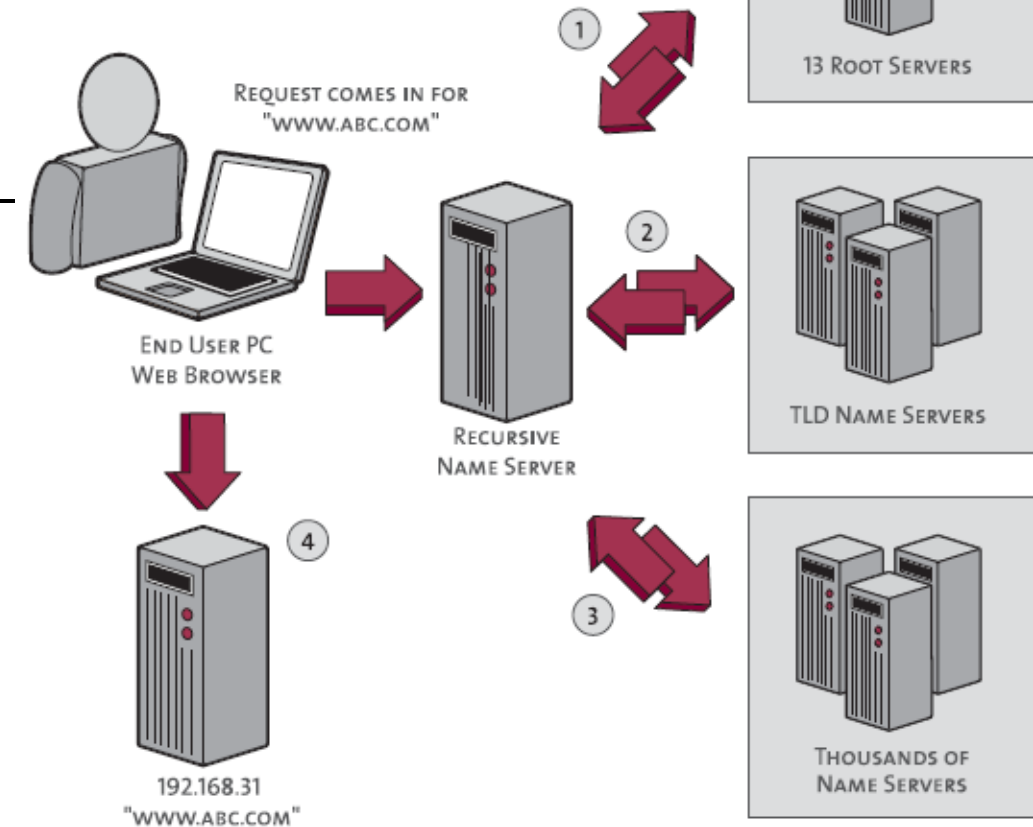
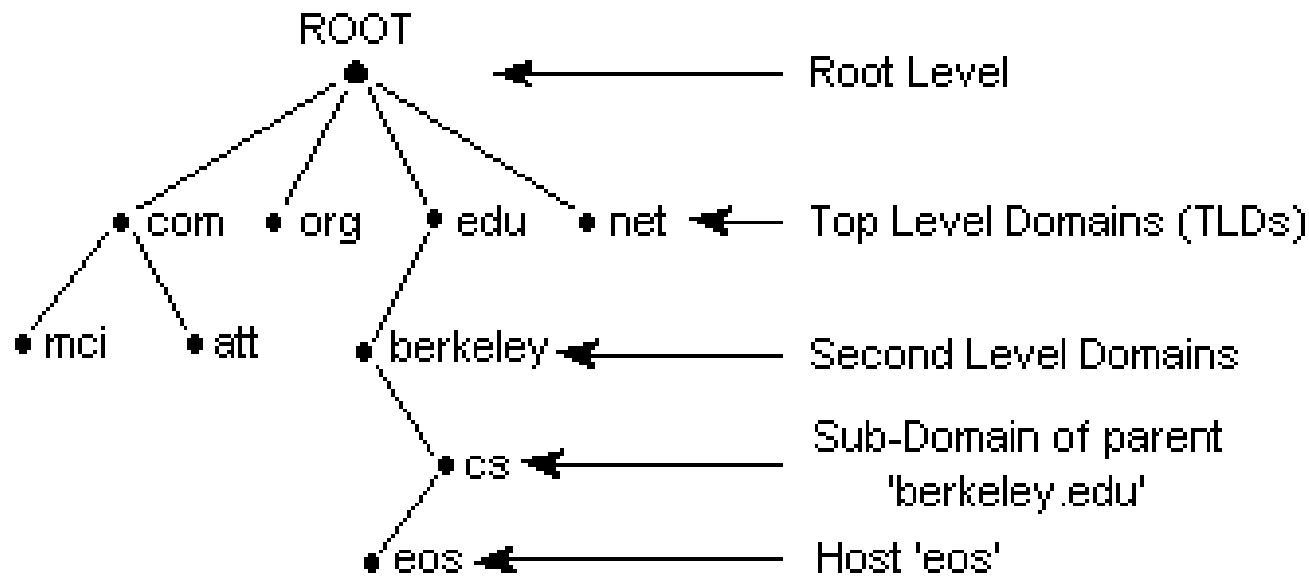
3. Now User can talk to google.com



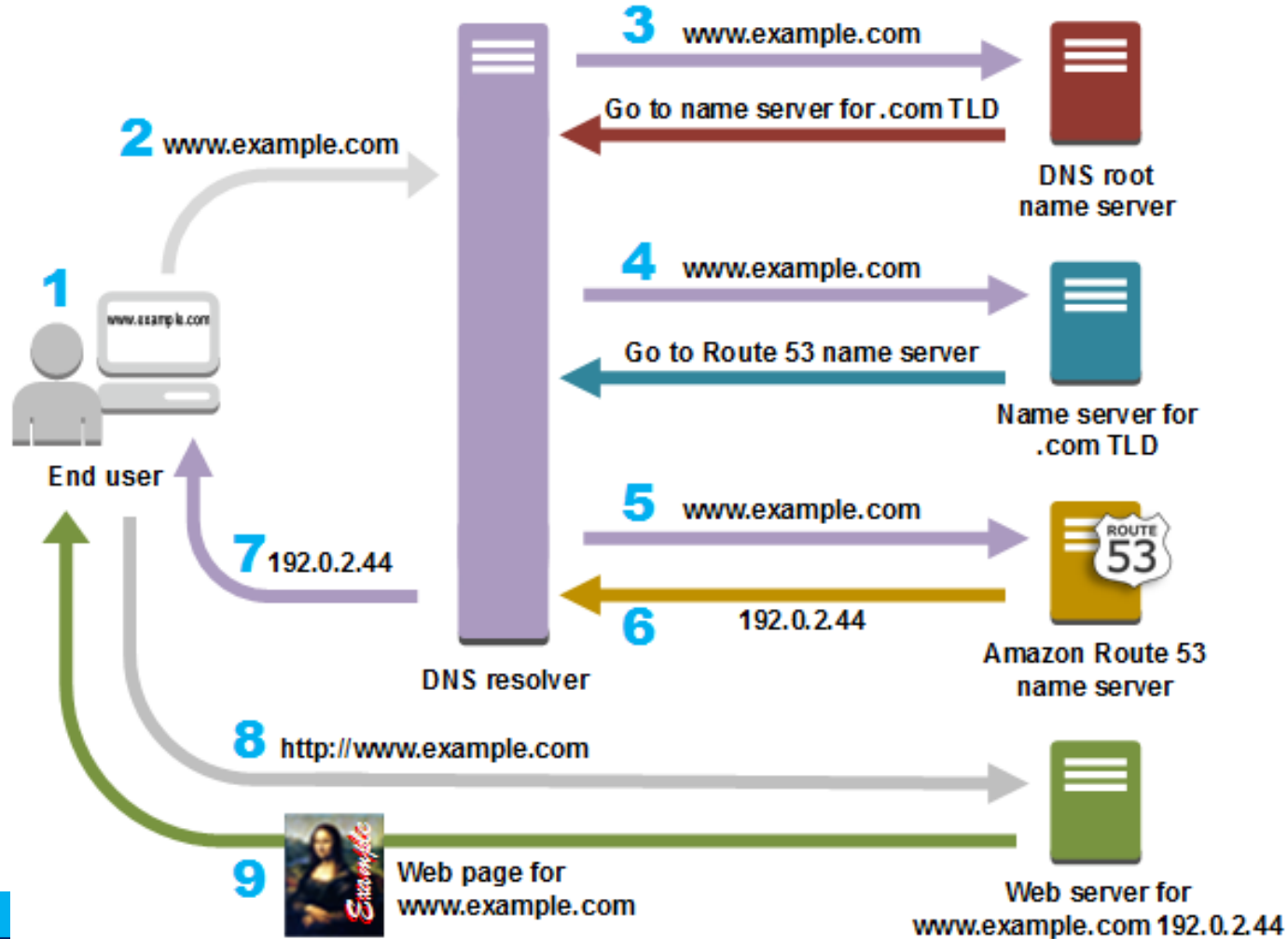


DNS is a hierarchical naming system

DNS Hierarchy



DNS Example



- DNS resolver is often managed by the user's Internet service provider
- TLD: top-level domain, such .com, org

Observe DNS with `dig`



dig

- It is a network administration tool used in Linux and other Unix-based operating systems to perform DNS (Domain Name System) queries.
 - retrieve information about DNS records, including
 - IP addresses associated with domain names,
 - mail server information, and
 - other DNS-related information.
- It sends a query to a specified DNS server and returns the response in a human-readable format.
 - It can be used to troubleshoot DNS issues, test DNS configuration changes, and gather information about DNS servers.



Create a working directory

```
(student@kalit01)-[~]  
$ mkdir dns  
  
(student@kalit01)-[~]  
$ cd dns
```



```
(student@kalit01)-[~/dns]
```

```
$ dig google.com
```

2. Generating dns traffic

```
; <<>> DiG 9.18.12-1-Debian <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 11464
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags::; udp: 65494
;; QUESTION SECTION:
;google.com.                IN      A

;; ANSWER SECTION:
google.com.                 25      IN      A      142.251.163.113
google.com.                 25      IN      A      142.251.163.139
google.com.                 25      IN      A      142.251.163.102
google.com.                 25      IN      A      142.251.163.100
google.com.                 25      IN      A      142.251.163.138
google.com.                 25      IN      A      142.251.163.101

;; Query time: 4 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Mon Mar 13 10:01:02 EDT 2023
;; MSG SIZE rcvd: 135
```

student@kalit01: ~/dns 105x7

```
(student@kalit01)-[~/dns]
```

```
$ tshark -i eth0 -w dig_dns.pcap
```

1. Capturing traffic

Capturing on 'eth0'

```
** (tshark:185262) 10:00:59.798575 [Main MESSAGE] -- Capture started.
** (tshark:185262) 10:00:59.798642 [Main MESSAGE] -- File: "dig_dns.pcap"
```

562 ^C 3. Control+c

tshark:



Icdfa

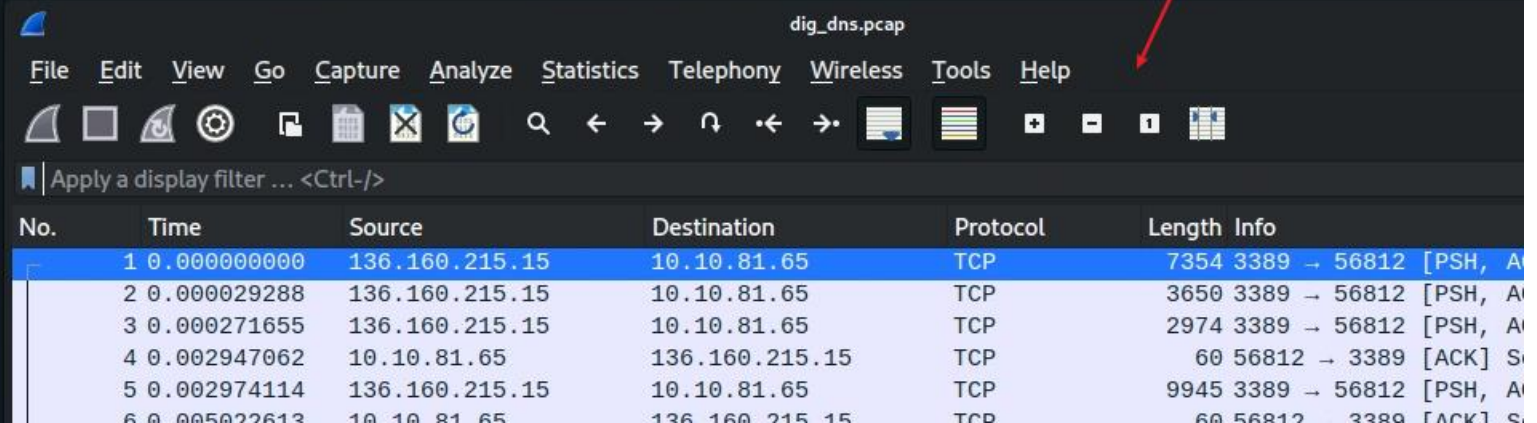
View captured DNS traffic

```
(student@kalit01)-[~/dns]  
$ ll
```

```
total 1764
```

```
-rw----- 1 student student 1804656 Mar 13 10:01 dig_dns.pcap
```

```
(student@kalit01)-[~/dns]  
$ wireshark dig_dns.pcap
```



dig_dns.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	136.160.215.15	10.10.81.65	TCP	7354	3389 → 56812 [PSH, A
2	0.000029288	136.160.215.15	10.10.81.65	TCP	3650	3389 → 56812 [PSH, A
3	0.000271655	136.160.215.15	10.10.81.65	TCP	2974	3389 → 56812 [PSH, A
4	0.002947062	10.10.81.65	136.160.215.15	TCP	60	56812 → 3389 [ACK] S
5	0.002974114	136.160.215.15	10.10.81.65	TCP	9945	3389 → 56812 [PSH, A
6	0.005022613	10.10.81.65	136.160.215.15	TCP	60	56812 → 3389 [ACK] S



wget https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Networking_Forensics/lab_files/dns/dig_dns.pcap



View layers DNS/UDP/IP/Frame of DNS query

FileEditViewGoCaptureAnalyzeStatisticsTelephonyWirelessToolsHelp

dns filter

dns

No.	Time	Source	Destination	Protocol	Length	Info
244	2.351280560	136.160.215.15	192.168.2.31	DNS	81	Standard query 0xb4a3 A google.com OPT
245	2.354063846	192.168.2.31	136.160.215.15	DNS	177	Standard query response 0xb4a3 A google.com A 142.251.163.113 A 142.251.163.139 A 142.251.163.141 A 142.251.163.142

▶ Frame 244: 81 bytes on wire (648 bits), 81 bytes captured (648 bits) on interface eth0, id 0

▶ Ethernet II, Src: VMware_86:cb:fc (00:50:56:86:cb:fc), Dst: PaloAlto_00:0a:30 (00:1b:17:00:0a:30)

▶ Internet Protocol Version 4, Src: 136.160.215.15, Dst: 192.168.2.31

▼ User Datagram Protocol, Src Port: 33039, Dst Port: 53

Source Port: 33039

Destination Port: 53

Length: 47

Checksum: 0x22b8 [unverified]

[Checksum Status: Unverified]

[Stream index: 0]

▶ [Timestamps]

UDP payload (39 bytes)

▼ Domain Name System (query)

Transaction ID: 0xb4a3

▶ Flags: 0x0100 Standard query

Questions: 1

Answer RRs: 0

Authority RRs: 0

Additional RRs: 1

▼ Queries

▼ google.com: type A, class IN

Name: google.com

[Name Length: 10]

[Label Count: 2]

Type: A (Host Address) (1)

Class: IN (0x0001)

▶ Additional records

[\[Response In: 245\]](#)

DNS use UDP

Query

0000 00 1b 17 00 0a 30 00 50 56 86 cb fc 08 00 45 00

0010 00 43 5b 2f 00 00 40 11 fd 03 88 a0 d7 0f c0 a0

0020 02 1f 81 0f 00 35 00 2f 22 b8 b4 a3 01 00 00 00

0030 00 00 00 00 00 01 06 67 6f 6f 67 6c 65 03 63 65

0040 6d 00 00 01 00 01 00 00 29 05 c0 00 00 00 00 00

0050 00

View DNS query source and destination IPs

dns

No.	Time	Source	Destination	Protocol	Length	Info
244	2.351280560	136.160.215.15	192.168.2.31	DNS	81	Standard query 0xb4a3 A google.com OPT
245	2.354063846	192.168.2.31	136.160.215.15	DNS	177	Standard query response 0xb4a3 A google.com

▶ Frame 244: 81 bytes on wire (648 bits), 81 bytes captured (648 bits) on interface eth0, id 0

▶ Ethernet II, Src: VMware_86:cb:fc (00:50:56:86:cb:fc), Dst: PaloAlto_00:0a:30 (00:1b:17:00:0a:30)

▼ Internet Protocol Version 4, Src: 136.160.215.15, Dst: 192.168.2.31

- 0100 = Version: 4
- 0101 = Header Length: 20 bytes (5)
- ▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
- Total Length: 67
- Identification: 0x5b2f (23343)
- ▶ 000. = Flags: 0x0
- ...0 0000 0000 0000 = Fragment Offset: 0
- Time to Live: 64
- Protocol: UDP (17)
- Header Checksum: 0xfd03 [validation disabled]
- [Header checksum status: Unverified]
- Source Address: 136.160.215.15
- Destination Address: 192.168.2.31

▶ User Datagram Protocol, Src Port: 33039, Dst Port: 53

▶ Domain Name System (query)

Kali

Local dns server



Local DNS configuration file

lists the DNS servers that your system is using for name resolution.

```
(student@kalit01)-[~/dns]  
$ cat /etc/resolv.conf  
# Generated by NetworkManager  
search lab.ubalt.edu  
nameserver 192.168.2.31  
nameserver 192.168.3.31
```

match DNS traffic



DNS response

dns

No.	Time	Source	Destination	Protocol	Length	Info
244	2.351280560	136.160.215.15	192.168.2.31	DNS	81	Standard query 0xb4a3 A google.com OPT
245	2.354063846	192.168.2.31	136.160.215.15	DNS	177	Standard query response 0xb4a3 A googl

▶ Frame 245: 177 bytes on wire (1416 bits), 177 bytes captured (1416 bits) on interface eth0, id 0
▶ Ethernet II, Src: PaloAlto_00:0a:30 (00:1b:17:00:0a:30), Dst: VMware_86:cb:fc (00:50:56:86:cb:fc)
▶ Internet Protocol Version 4, Src: 192.168.2.31, Dst: 136.160.215.15
▶ User Datagram Protocol, Src Port: 53, Dst Port: 33039
▼ Domain Name System (response)
Transaction ID: 0xb4a3
▶ Flags: 0x8180 Standard query response, No error
Questions: 1
Answer RRs: 6
Authority RRs: 0
Additional RRs: 1
▼ Queries
▼ google.com: type A, class IN
Name: google.com
[Name Length: 10]
[Label Count: 2]
Type: A (Host Address) (1)
Class: IN (0x0001)
▼ Answers
▶ google.com: type A, class IN, addr 142.251.163.113
▶ google.com: type A, class IN, addr 142.251.163.139
▶ google.com: type A, class IN, addr 142.251.163.102
▶ google.com: type A, class IN, addr 142.251.163.100
▶ google.com: type A, class IN, addr 142.251.163.138
▶ google.com: type A, class IN, addr 142.251.163.101
▶ Additional records
[Request In: 244]
[Time: 0.002783286 seconds]

DNS response

Answers

One of answers

```
▼ Domain Name System (response)
  Transaction ID: 0xb4a3
  ▶ Flags: 0x8180 Standard query response, No error
  Questions: 1
  Answer RRs: 6
  Authority RRs: 0
  Additional RRs: 1
  ▼ Queries
    ▼ google.com: type A, class IN
      Name: google.com
      [Name Length: 10]
      [Label Count: 2]
      Type: A (Host Address) (1)
      Class: IN (0x0001)
  ▼ Answers
    ▼ google.com: type A, class IN, addr 142.251.163.113
      Name: google.com
      Type: A (Host Address) (1)
      Class: IN (0x0001)
      Time to live: 25 (25 seconds)
      Data length: 4
      Address: 142.251.163.113
    ▶ google.com: type A, class IN, addr 142.251.163.139
    ▶ google.com: type A, class IN, addr 142.251.163.102
    ▶ google.com: type A, class IN, addr 142.251.163.100
    ▶ google.com: type A, class IN, addr 142.251.163.138
    ▶ google.com: type A, class IN, addr 142.251.163.101
  ▶ Additional records
    [Request In: 244]
    [Time: 0.002783286 seconds]
```

- maps the domain name to an IP address.
- refers to the Internet class.



Observe the DNS traffic of accessing google.com

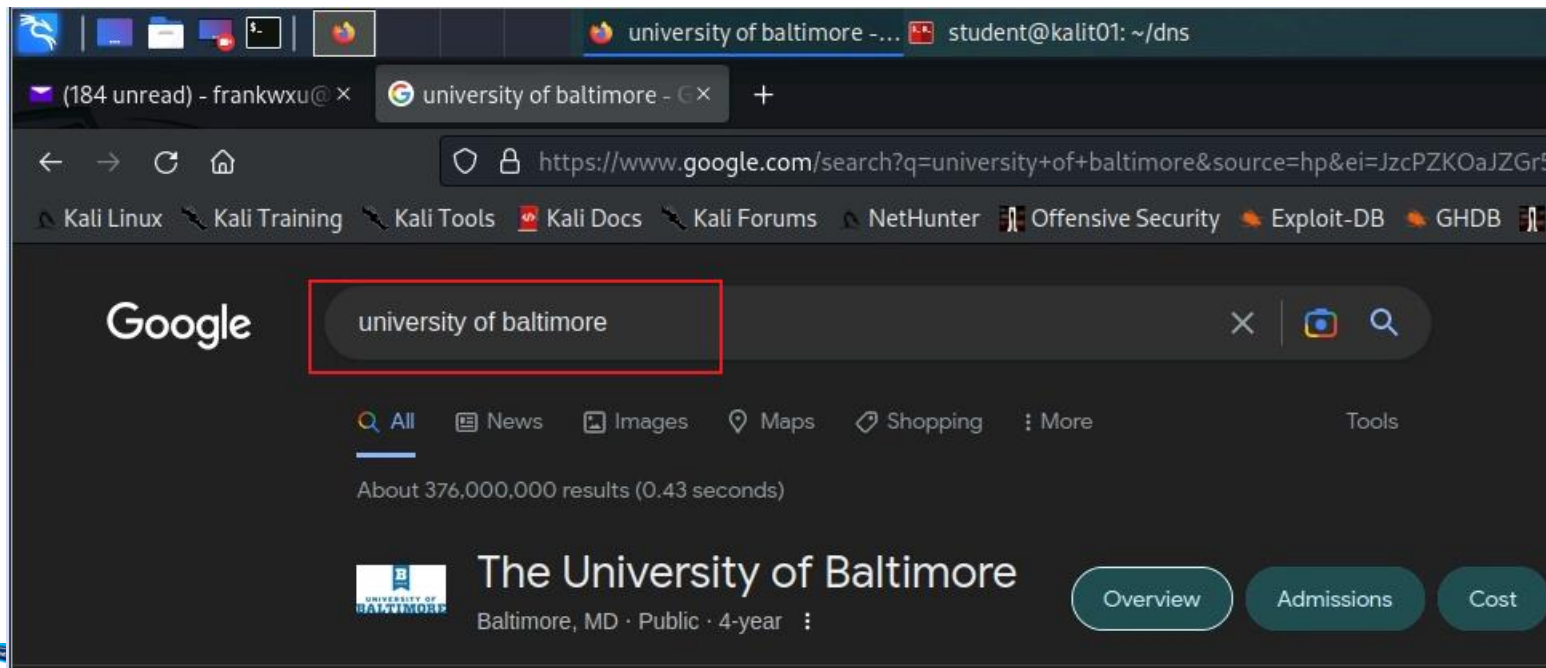
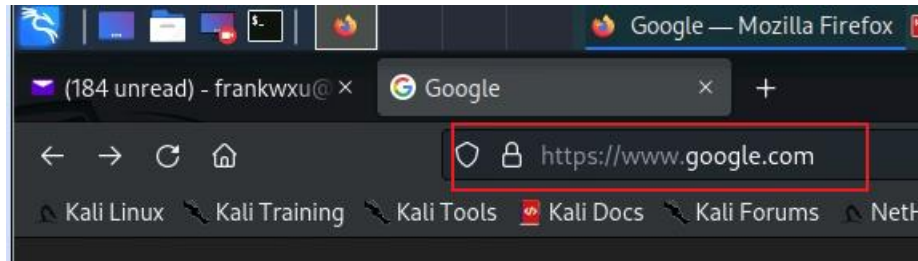


Command to capture traffic

```
student@kalit01: ~/dns 105x17
(student@kalit01)-[~/dns]
$ tshark -i eth0 -w http_dns.pcap
Capturing on 'eth0'
** (tshark:186710) 10:45:52.135152 [Main MESSAGE] -- Capture started.
** (tshark:186710) 10:45:52.135310 [Main MESSAGE] -- File: "http_dns.pcap"
6897 ^C
tshark:
```



Accessing google.com and google search



Multiple pairs of DNS query and response because Google results page contains multiple hidden URLs (ads, images)

No.	Time	Source	Destination	Protocol	Length	Info
448	7.001331416	136.160.215.15	192.168.2.31	DNS	85	Standard query 0x0b4c A www.google.com OPT
449	7.004296880	192.168.2.31	136.160.215.15	DNS	181	Standard query response 0x0b4c A www.google.com A 142.251.163.106 A 142.251.163.104 A 142.251.163.102
508	7.168967105	136.160.215.15	192.168.2.31	DNS	86	Standard query 0xc6c6 A www.gstatic.com OPT
509	7.171260202	192.168.2.31	136.160.215.15	DNS	102	Standard query response 0xc6c6 A www.gstatic.com A 172.253.63.94 OPT
877	7.597732573	136.160.215.15	192.168.2.31	DNS	86	Standard query 0x1fb2 A apis.google.com OPT
878	7.600481169	192.168.2.31	136.160.215.15	DNS	203	Standard query response 0x1fb2 A apis.google.com CNAME plus.l.google.com A 172.253.115.101
885	7.613979796	136.160.215.15	192.168.2.31	DNS	85	Standard query 0xb37e A ogs.google.com OPT
891	7.616809262	192.168.2.31	136.160.215.15	DNS	202	Standard query response 0xb37e A ogs.google.com CNAME www3.l.google.com A 142.251.163.139 A 142.251.163.137
1044	7.741376643	136.160.215.15	192.168.2.31	DNS	88	Standard query 0x788d A fonts.gstatic.com OPT
1046	7.743667806	192.168.2.31	136.160.215.15	DNS	104	Standard query response 0x788d A fonts.gstatic.com A 172.253.122.94 OPT
1292	8.032640338	136.160.215.15	192.168.2.31	DNS	91	Standard query 0xe579 A adservice.google.com OPT
1294	8.035004001	192.168.2.31	136.160.215.15	DNS	123	Standard query response 0xe579 A adservice.google.com A 172.253.63.154 A 172.253.63.155 A 172.253.63.156
1311	8.055027168	136.160.215.15	192.168.2.31	DNS	86	Standard query 0xff44 A play.google.com OPT
1312	8.057694788	192.168.2.31	136.160.215.15	DNS	182	Standard query response 0xff44 A play.google.com A 142.251.163.101 A 142.251.163.139 A 142.251.163.102
1381	8.205834328	136.160.215.15	192.168.2.31	DNS	90	Standard query 0x10ab A googleads.g.doubleclick.net OPT
1383	8.208152910	192.168.2.31	136.160.215.15	DNS	130	Standard query response 0x10ab A googleads.g.doubleclick.net A 172.253.62.157 A 172.253.62.158 A 172.253.62.159
1388	8.214932976	136.160.215.15	192.168.2.31	DNS	98	Standard query 0xa54d A safebrowsing.googleapis.com OPT
1389	8.217530617	192.168.2.31	136.160.215.15	DNS	114	Standard query response 0xa54d A safebrowsing.googleapis.com A 142.251.16.95 OPT
2025	17.716448151	136.160.215.15	192.168.2.31	DNS	97	Standard query 0xb5df A encrypted-tbn0.gstatic.com OPT
2026	17.719280345	192.168.2.31	136.160.215.15	DNS	193	Standard query response 0xb5df A encrypted-tbn0.gstatic.com A 172.253.115.113 A 172.253.115.114 A 172.253.115.115
2512	21.168919125	136.160.215.15	192.168.2.31	DNS	96	Standard query 0xc230 A lh5.googleusercontent.com OPT
2516	21.171259181	192.168.2.31	136.160.215.15	DNS	141	Standard query response 0xc230 A lh5.googleusercontent.com CNAME googlehosted.l.googleusercontent.com A 142.251.16.94
2542	21.194041420	136.160.215.15	192.168.2.31	DNS	84	Standard query 0x1ad7 A ocsdp.ki.goog OPT
2543	21.196389535	192.168.2.31	136.160.215.15	DNS	135	Standard query response 0x1ad7 A ocsdp.ki.goog CNAME pki-goog.l.google.com A 142.251.16.94
4248	31.819012899	136.160.215.15	192.168.2.31	DNS	84	Standard query 0x111f A id.google.com OPT
4249	31.821454919	192.168.2.31	136.160.215.15	DNS	100	Standard query response 0x111f A id.google.com A 172.217.30.195 OPT

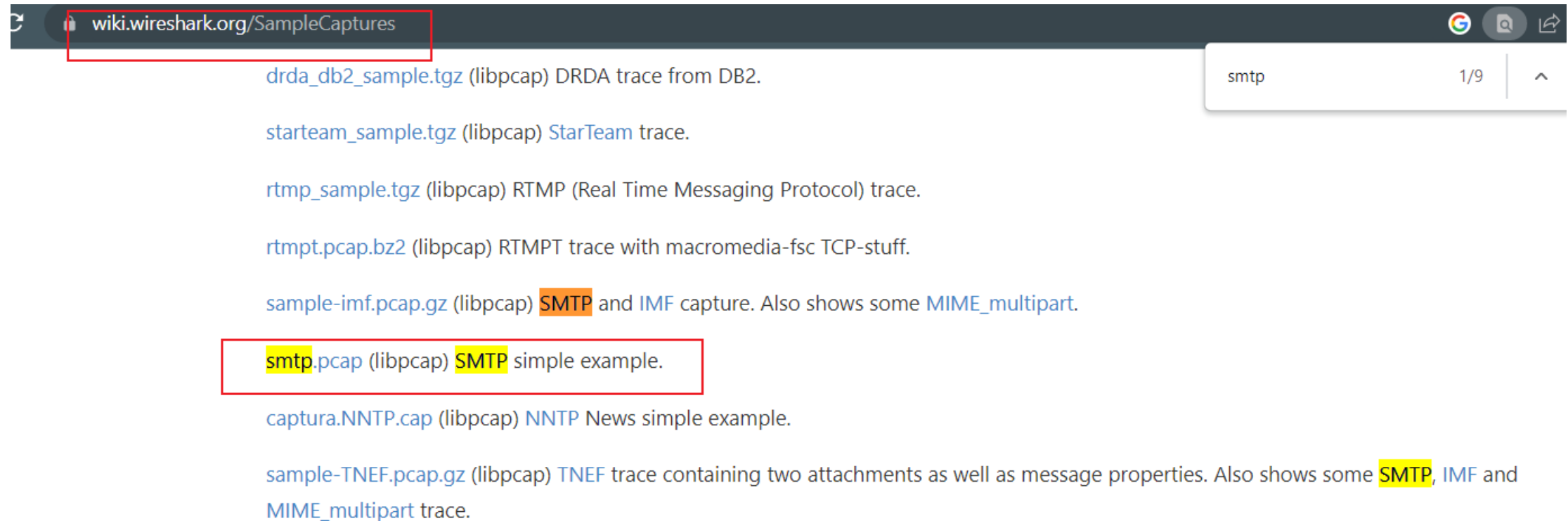
Multiple pairs of query and response



DNS analysis of Email traffic



Lab Files



The screenshot shows a web browser window with the address bar containing wiki.wireshark.org/SampleCaptures. A search bar on the right contains the text "smtp" and shows "1/9" results. The main content area lists several sample capture files:

- [drda_db2_sample.tgz](#) (libpcap) DRDA trace from DB2.
- [starteam_sample.tgz](#) (libpcap) [StarTeam](#) trace.
- [rtmp_sample.tgz](#) (libpcap) RTMP (Real Time Messaging Protocol) trace.
- [rtmpt.pcap.bz2](#) (libpcap) RTMPT trace with macromedia-fsc TCP-stuff.
- [sample-imf.pcap.gz](#) (libpcap) **SMTP** and **IMF** capture. Also shows some [MIME_multipart](#).
- [smtp.pcap](#)** (libpcap) **SMTP** simple example.
- [captura.NNTP.cap](#) (libpcap) [NNTP](#) News simple example.
- [sample-TNEF.pcap.gz](#) (libpcap) [TNEF](#) trace containing two attachments as well as message properties. Also shows some **SMTP**, **IMF** and [MIME_multipart](#) trace.

<https://wiki.wireshark.org/SampleCaptures>



Download smtp.pcap

```
(kali㉿kali)-[~/smtp]
└─$ wget https://wiki.wireshark.org/uploads/__moin_import__/attachments/SampleCaptures/smtp.pcap
--2023-03-10 14:58:08-- https://wiki.wireshark.org/uploads/__moin_import__/attachments/SampleCaptures/smtp.pcap
Resolving wiki.wireshark.org (wiki.wireshark.org)... 172.67.75.39, 104.26.10.240, 104.26.11.240, ...
Connecting to wiki.wireshark.org (wiki.wireshark.org)|172.67.75.39|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 27850 (27K) [application/octet-stream]
Saving to: 'smtp.pcap'

smtp.pcap          100%[=====>]  27.20K  --.-KB/s    in 0.004s

2023-03-10 14:58:08 (6.91 MB/s) - 'smtp.pcap' saved [27850/27850]
```



DNS query

Router please find the IP of **mail.patriots.in**

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.10.1.4	10.10.1.1	DNS	76	Standard query 0x7956 A mail.patriots.in
2	0.034025	10.10.1.1	10.10.1.4	DNS	142	Standard query response 0x7956 A mail.patriots.in
3	0.036986	10.10.1.4	74.53.140.153	TCP	62	1470 → 25 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SAC
4	0.383936	74.53.140.153	10.10.1.4	TCP	62	25 → 1470 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MS
5	0.383968	10.10.1.4	74.53.140.153	TCP	54	1470 → 25 [ACK] Seq=1 Ack=1 Win=65535 Len=0
6	0.727603	74.53.140.153	10.10.1.4	SMTP	235	S: 220-xc90.websitewelcome.com ESMTP Exim 4.69 #1
7	0.732749	10.10.1.4	74.53.140.153	SMTP	63	C: EHLO GP
8	1.073326	74.53.140.153	10.10.1.4	TCP	60	25 → 1470 [ACK] Seq=182 Ack=10 Win=5840 Len=0
9	1.074123	74.53.140.153	10.10.1.4	SMTP	191	S: 250-xc90.websitewelcome.com Hello GP [122.162.1
10	1.076669	10.10.1.4	74.53.140.153	SMTP	66	C: AUTH LOGIN

▶ Frame 1: 76 bytes on wire (608 bits), 76 bytes captured (608 bits)
▶ Ethernet II, Src: Cradlepo_3c:17:c2 (00:e0:1c:3c:17:c2), Dst: Netgear_d9:81:60 (00:1f:33:d9:81:60)
▶ Internet Protocol Version 4, Src: 10.10.1.4, Dst: 10.10.1.1
▶ User Datagram Protocol, Src Port: 56166, Dst Port: 53

▼ Domain Name System (query)
Transaction ID: 0x7956
▶ Flags: 0x0100 Standard query
Questions: 1
Answer RRs: 0
Authority RRs: 0
Additional RRs: 0
▼ Queries
▶ mail.patriots.in: type A, class IN
[Response in: 2]

Application layer
protocol

asking it to return the IPv4 (**Type A**) address
associated with a given domain name, using
the Internet namespace (**Class IN**).



DNS query response

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.10.1.4	10.10.1.1	DNS	76	Standard query 0x7956 A mail.patriots.in
2	0.034025	10.10.1.1	10.10.1.4	DNS	142	Standard query response 0x7956 A mail.patriots.in
3	0.036986	10.10.1.4	74.53.140.153	TCP	62	1470 → 25 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SAQ
4	0.383936	74.53.140.153	10.10.1.4	TCP	62	25 → 1470 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MS
5	0.383968	10.10.1.4	74.53.140.153	TCP	54	1470 → 25 [ACK] Seq=1 Ack=1 Win=65535 Len=0
6	0.727603	74.53.140.153	10.10.1.4	SMTP	235	S: 220-xc90.websitewelcome.com ESMTP Exim 4.69 #1
7	0.732749	10.10.1.4	74.53.140.153	SMTP	63	C: EHLO GP
8	1.073326	74.53.140.153	10.10.1.4	TCP	60	25 → 1470 [ACK] Seq=182 Ack=10 Win=5840 Len=0
9	1.074123	74.53.140.153	10.10.1.4	SMTP	191	S: 250-xc90.websitewelcome.com Hello GP [122.162.1
10	1.076669	10.10.1.4	74.53.140.153	SMTP	66	C: AUTH LOGIN

▶ Frame 2: 142 bytes on wire (1136 bits), 142 bytes captured (1136 bits)

▶ Ethernet II, Src: Netgear_d9:81:60 (00:1f:33:d9:81:60), Dst: Cradlepo_3c:17:c2 (00:e0:1c:3c:17:c2)

▶ Internet Protocol Version 4, Src: 10.10.1.1, Dst: 10.10.1.4

▶ User Datagram Protocol, Src Port: 53, Dst Port: 56166

▼ Domain Name System (response)

Transaction ID: 0x7956

▶ Flags: 0x8180 Standard query response, No error

Questions: 1

Answer RRs: 2

Authority RRs: 2

Additional RRs: 0

▼ Queries

▶ mail.patriots.in: type A, class IN

▼ Answers

▶ mail.patriots.in: type CNAME, class IN, cname patriots.in

▶ patriots.in: type A, class IN, addr 74.53.140.153

▼ Authoritative nameservers

▶ patriots.in: type NS, class IN, ns ns2.patriots.in

▶ patriots.in: type NS, class IN, ns ns1.patriots.in

[Request In: 1]

[Time: 0.034025000 seconds]

Answer to the DNS query



Authoritative nameservers = authoritative DNS servers

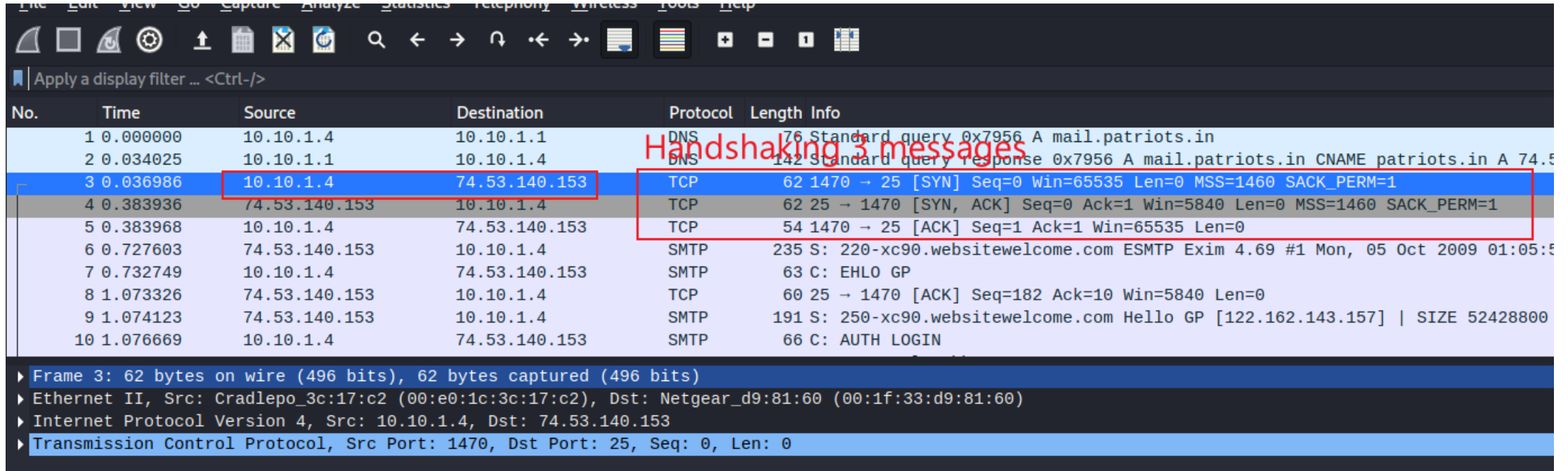
```
▼ Answers
  ▶ mail.patriots.in: type CNAME, class IN, cname patriots.in
  ▶ patriots.in: type A, class IN, addr 74.53.140.153
  ▼ Authoritative nameservers
    ▼ patriots.in: type NS, class IN, ns ns2.patriots.in
      Name: patriots.in
      Type: NS (authoritative Name Server) (2)
      Class: IN (0x0001)
      Time to live: 82828 (23 hours, 28 seconds)
      Data length: 6
      Name Server: ns2.patriots.in
    ▼ patriots.in: type NS, class IN, ns ns1.patriots.in
      Name: patriots.in
      Type: NS (authoritative Name Server) (2)
      Class: IN (0x0001)
      Time to live: 82828 (23 hours, 28 seconds)
      Data length: 6
      Name Server: ns1.patriots.in
[Request In: 1]
[Time: 0.034025000 seconds]

0000 00 e0 1c 3c 17 c2 00 1f 33 d9 81 60 08 00 45 00 ...<... 3...`..E.
0010 00 80 00 00 40 00 40 11 24 55 0a 0a 01 01 0a 0a ...@.@.$U.....
0020 01 04 00 35 db 66 00 6c 2d 2d 79 56 81 80 00 01 ...5.f.l--yV....
0030 00 02 00 02 00 00 04 6d 61 69 6c 08 70 61 74 72 ...m ail patr
0040 69 6f 74 73 02 69 6e 00 00 01 00 01 c0 0c 00 05 iots.in .....
0050 00 01 00 00 2a 4b 00 02 c0 11 c0 11 00 01 00 01 ...*K.....
0060 00 00 2a 4c 00 04 4a 35 8c 99 c0 11 00 02 00 01 ...*L..J5.....
0070 00 01 43 8c 00 06 03 6e 73 32 c0 11 c0 11 00 02 ...C....n s2.....
0080 00 01 00 01 43 8c 00 06 03 6e 73 31 c0 11 ...C... ns1..
```

- Answer from the router DNS caching
- The original and official information about a domain name from ns1.patriots.in or ns1.patriots.in.
- The DNS entry will expire
- Possible DNS attack



Handshaking (SYN, [SYN, ACK], ACK)



Handshaking 3 messages

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.10.1.4	10.10.1.1	DNS	76	Standard query 0x7956 A mail.patriots.in
2	0.034025	10.10.1.1	10.10.1.4	DNS	142	Standard query response 0x7956 A mail.patriots.in CNAME patriots.in A 74.53.140.153
3	0.036986	10.10.1.4	74.53.140.153	TCP	62	1470 → 25 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM=1
4	0.383936	74.53.140.153	10.10.1.4	TCP	62	25 → 1470 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1460 SACK_PERM=1
5	0.383968	10.10.1.4	74.53.140.153	TCP	54	1470 → 25 [ACK] Seq=1 Ack=1 Win=65535 Len=0
6	0.727603	74.53.140.153	10.10.1.4	SMTP	235	S: 220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:5
7	0.732749	10.10.1.4	74.53.140.153	SMTP	63	C: EHLO GP
8	1.073326	74.53.140.153	10.10.1.4	TCP	60	25 → 1470 [ACK] Seq=182 Ack=10 Win=5840 Len=0
9	1.074123	74.53.140.153	10.10.1.4	SMTP	191	S: 250-xc90.websitewelcome.com Hello GP [122.162.143.157] SIZE 52428800
10	1.076669	10.10.1.4	74.53.140.153	SMTP	66	C: AUTH LOGIN

▶ Frame 3: 62 bytes on wire (496 bits), 62 bytes captured (496 bits)

▶ Ethernet II, Src: Cradlepo_3c:17:c2 (00:e0:1c:3c:17:c2), Dst: Netgear_d9:81:60 (00:1f:33:d9:81:60)

▶ Internet Protocol Version 4, Src: 10.10.1.4, Dst: 74.53.140.153

▶ Transmission Control Protocol, Src Port: 1470, Dst Port: 25, Seq: 0, Len: 0

